
Discrete Codebooks and Temporal Context: Bridging the Non-Markovian Gap in Pac-Man World Models

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Abstract

We investigate the trade-off between continuous and discrete latent spaces for a multitask world model on Ms. Pac-Man, evaluating five architectures: a continuous autoencoder, a VQ-VAE, a Temporal Transformer for non-Markovian danger prediction, and two generative priors (token-level and frame-level). On a 10k-frame dataset with 94%/6% class imbalance, the discrete VQ-VAE achieves 96.25 ± 0.23 accuracy (+4.23 pp over the continuous baseline); 8-frame temporal aggregation adds another +2.07 pp. A 15-step PSNR rollout exposes complementary failure modes: the token-level prior collapses to 14.33 dB at step 1, while the frame-level prior holds at 30.79 dB after 15 steps but biases moving agents toward the conditional mean.

1. Introduction

World models compress environment dynamics into compact latent representations for downstream planning (Ha & Schmidhuber, 2018). Long-horizon simulation is constrained by the choice of latent space: continuous embeddings blur moving entities under deterministic MSE objectives, while discrete tokenization is exposed to error accumulation during sequential generation.

We study this continuous-discrete trade-off in a multitask regime (Caruana, 1997) on Ms. Pac-Man (Bellemare et al., 2013; Towers et al., 2023). The model performs simultaneous frame reconstruction and look-ahead *danger prediction*: each frame is labelled by whether a life will be lost within a short future horizon. This non-Markovian task requires predicting future penalties from visually safe states, necessitating explicit temporal modeling.

To dissect this trade-off, we evaluate a progression of five

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models, each motivated by a limitation of the previous one. Two multitask autoencoders share a reconstruction-plus-classification objective: a continuous Baseline establishes the reference, and a VQ-VAE replaces its unstructured latent with a discrete codebook. The remaining three are single-task heads built on the frozen VQ-VAE: a Temporal Transformer aggregates 8-frame context to address the non-Markovian label, while token-level and frame-level priors probe the discrete/continuous trade-off in autoregressive future simulation. Code and weights are available at <https://github.com/leonardoCosta02/pacman-world-model>.

2. Related Work

Early world models (Ha & Schmidhuber, 2018) relied on continuous autoencoders (Kingma & Welling, 2014), which suffer from compounding blurriness during autoregressive rollouts. To mitigate this, vector quantization (van den Oord et al., 2017) and transformer-based transition dynamics (Esser et al., 2021) shifted the paradigm towards discrete codebooks. This discrete-autoregressive synergy achieves state-of-the-art sample efficiency in Atari (Micheli et al., 2023) and robust continuous control (Hafner et al., 2023). We systematically benchmark these latent topologies against a continuous baseline, additionally assessing representation robustness via magnitude-based pruning (Han et al., 2015; Frankle & Carbin, 2019).

3. Methodology

Our pipeline decouples spatial compression, temporal aggregation, and generative simulation.

Dataset and labelling. We collect 80×80 grayscale Ms. Pac-Man frames via a random policy, yielding a 10k-frame dataset for feature extraction and a 50k-frame dataset for generative rollouts. We define a non-Markovian danger prediction task: observation o_t receives label $y_t = 1$ (DANGER) if a life is lost in $t' \in (t, t + 15]$. This induces a severe 94%/6% class imbalance.

Spatial representation. The continuous baseline is a 3-layer CNN mapping to a 6400-dimensional latent, optimized for reconstruction and classification ($\mathcal{L} = 10\mathcal{L}_{\text{MSE}} + \mathcal{L}_{\text{BCE}}$) with a 15:1 positive class weight. The discrete VQ-VAE replaces the bottleneck with a 4-layer encoder, quantizing o_t into a 10×10 grid over $K = 128$ tokens ($d = 64$). We optimize the codebook via exponential moving average ($\gamma = 0.99$) (van den Oord et al., 2017) to prevent index collapse. The loss balances reconstruction, classification, and commitment ($\alpha = 25, \beta = 1$).

Temporal aggregation and pruning. A single frozen VQ-VAE grid cannot reveal motion direction, so a Temporal Transformer (Vaswani et al., 2017) (4 layers, 4 heads, $d_{\text{model}} = 128$) processes 8-frame histories to recover the velocity cues the non-Markovian label depends on. Classification is anchored to a learnable [CLS] token (Devlin et al., 2019) to aggregate sequence logits. To evaluate representation redundancy, we apply 20% unstructured L_1 magnitude pruning (Han et al., 2015) to the VQ-VAE encoder and evaluate downstream classification zero-shot.

Generative priors. We train two state-transition models on the 50k dataset; the frame-level prior was introduced to address the failure mode of the token-level one. The discrete token-level prior is a causal Transformer decoder (4 layers, 4 heads, $d_{\text{model}} = 256$) predicting 800 flattened tokens autoregressively via top- k sampling. To mitigate exposure bias, training combines teacher forcing with noise injection (probability 0.15, decaying over the first 60% of training) and scheduled sampling (Bengio et al., 2015) (ramped from 0 to 0.40 from epoch 5). The continuous frame-level prior is a causal Transformer encoder (4 layers, 8 heads, $d_{\text{model}} = 256$) operating seq-to-one: it predicts the next $64 \times 10 \times 10$ unquantized latent volume from the 8-frame context in a single step via MSE optimization.

4. Experimental Results

We evaluate the core classification architectures across three independent random seeds to isolate representation topology from initialization variance (Table 1).

Table 1. Test-set performance over 3 random seeds (mean $\pm$ std).

Model	Accuracy (%)	F1 (DANGER)
Baseline (continuous)	92.02 $\pm$ 1.13	0.60 $\pm$ 0.03
VQ-VAE (discrete)	96.25 $\pm$ 0.23	0.76 $\pm$ 0.01
Temporal Transformer	98.32 $\pm$ 0.78	0.87 $\pm$ 0.06
VQ-VAE + 20% L1 pruning	96.38 $\pm$ 0.60	0.76 $\pm$ 0.03

Classification and pruning. Discretizing the latent space improves mean accuracy by +4.23 and F1 by +0.16 over the continuous baseline. Temporal sequence modeling via

the Transformer yields the highest performance by reducing non-Markovian false positives. Evaluating the downstream classifier zero-shot on representations from a 20% L_1 -pruned VQ-VAE yields $\Delta = +0.13 \pm 0.65$ pp across seeds, confirming that pruning causes no statistically significant degradation. The pruning mask disproportionately targets the deepest convolutional layer (23% sparsity) rather than early edge-detectors (2.9%), demonstrating high redundancy in the discrete topological mapping.

Latent geometry. All 128 codewords are active on the test set, confirming that the EMA update prevents codebook collapse. Linear interpolation between distant SAFE and DANGER latents produces a smooth, monotonic transition in the continuous baseline; the same interpolation in the VQ-VAE exhibits abrupt jumps, since each intermediate vector is re-quantised to the nearest codeword.

Autoregressive rollouts. We evaluate future simulation by measuring Peak Signal-to-Noise Ratio (PSNR) over 15 autoregressive steps across 50 unobserved contexts. The token-level prior (100 sampling steps per frame) degrades instantly, dropping to 14.33 ± 0.12 dB by step 1. The continuous frame-level prior generates full latents in a single pass, maintaining geometric coherence and stabilizing at 30.79 ± 2.98 dB after 15 steps. Visual inspection reveals complementary failure modes. The frame-level prior’s MSE objective biases moving entities toward the conditional mean; the re-quantisation step then maps these averaged features to background codewords, so Pac-Man and the ghosts fade after a few steps while the maze structure is preserved. The token-level prior fails in the opposite way: its local entity identity survives at the codeword level, but the spatial geometry across the 10×10 grid degrades immediately.

Inference latency. The Temporal Transformer processes 8-frame histories in 1.45 ms on an NVIDIA T4 GPU (690 sequences/s), satisfying 60 FPS real-time constraints.

5. Conclusion

Three observations follow. First, on this multitask objective a discrete codebook is statistically superior to a continuous bottleneck of comparable size (van den Oord et al., 2017; Micheli et al., 2023; Hafner et al., 2023). Second, an 8-frame Transformer with a learned [CLS] token bridges the gap between single-frame views and the predictive look-ahead label. Third, the two priors trade off cleanly: categorical autoregression preserves local entity identity but drifts geometrically, while continuous MSE prediction keeps structure but vanishes moving entities via regression to the mean. Limitations include the random data-collection policy and the absence of closed-loop control.

Statement on the Use of AI

In accordance with course guidelines, Google’s Gemini was used as a coding and writing assistant. Specifically, the tool was used to: (i) generate boilerplate PyTorch code and Hydra configurations for the five evaluated architectures; (ii) assist in debugging tensor shape mismatches during Kaggle deployments; and (iii) correct grammar and improve the conciseness of this manuscript.

I exclusively conceived all core design decisions, including model selection, the non-Markovian labelling scheme, the pruning experiments, and the formulation of the generative priors. All numerical results, multi-seed metrics, and PSNR evaluations were generated from independent executions on Kaggle GPUs and verified against the released checkpoints. No experimental data was fabricated. I fully understand the codebase and take complete responsibility for the submitted work.

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Appendix

A. Autoregressive rollout quality (PSNR). Figure 1 reports the per-step PSNR of the two priors against ground-truth frames, averaged over 50 unobserved contexts; error bars denote one standard deviation across contexts. The token-level prior collapses to ~ 14 dB already at step 1 and stays flat, while the frame-level prior starts near 34 dB and degrades gracefully to 30.79 ± 2.98 dB at step 15. The qualitative failure modes underlying these curves (geometric drift vs. entity vanishing) are discussed in Section 4.

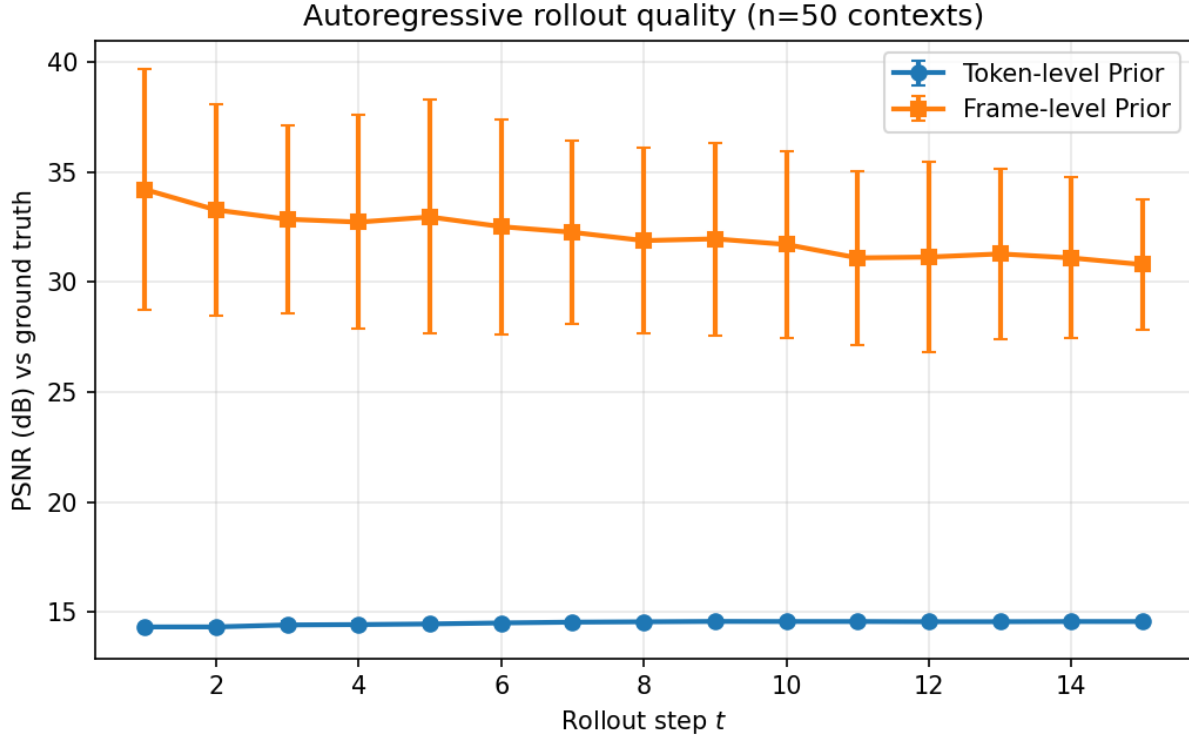


Figure 1. Autoregressive rollout quality: per-step PSNR (dB) vs. ground truth over 15 steps, averaged across 50 contexts. The token-level prior (blue) collapses at step 1; the frame-level prior (orange) holds above 30 dB.

B. Qualitative rollout comparison. Figure 2 shows the first five generated frames for both priors. The token-level prior produces structured noise from the very first generated frame, whereas the frame-level prior preserves the maze structure throughout the rollout. The frame-level prior is also about $100\times$ faster, since it generates each frame in a single forward pass rather than sampling 100 tokens sequentially.

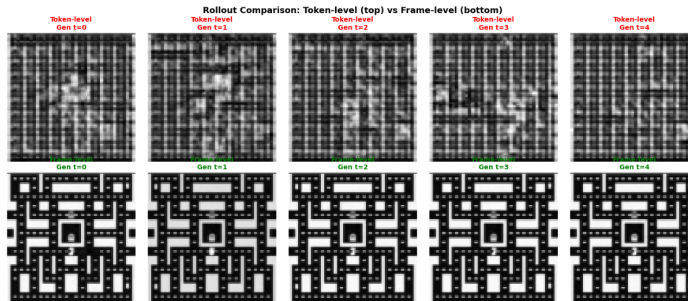


Figure 2. Rollout comparison, first five generated frames: token-level prior (top) vs. frame-level prior (bottom). The token-level rollout is structured noise from the first frame, while the frame-level rollout keeps the maze layout intact.

C. Architectural specifications. The continuous baseline encoder is a 3-layer CNN (channels: $1 \rightarrow 32 \rightarrow 64 \rightarrow 64$; kernel: 4×4 ; stride: 2) mapping inputs to a $10 \times 10 \times 64$ tensor, flattened to a 6400-dimensional latent. The decoder is symmetric, using transposed convolutions and a Sigmoid output. We note that the Baseline contains a redundant 6400×6400 linear layer between the encoder and decoder ($\sim 41\text{M}$ of its $\sim 42\text{M}$ parameters); a plain tensor reshape would suffice, reducing the effective parameter count to $\sim 1\text{M}$, comparable to the VQ-VAE. We report it as implemented, since all results were obtained with this architecture. The VQ-VAE shares this backbone but adds a 1×1 convolution to project features to $d = 64$. The codebook contains $K = 128$ vectors, optimized via EMA ($\gamma = 0.99$, commitment cost 1.0). The Temporal Transformer processes 8-frame sequences using a 4-layer, 4-head encoder ($d_{\text{model}} = 128$, FFN dimension 512). The token-level prior is a causal 4-layer, 4-head Transformer decoder ($d_{\text{model}} = 256$). The frame-level prior is a 4-layer, 8-head causal Transformer encoder ($d_{\text{model}} = 256$) used in seq-to-one mode.

D. Optimization protocols. All models are trained using Adam. The baseline autoencoder is trained for 10 epochs ($\text{lr}=10^{-3}$, weight decay= 10^{-4} , $\alpha = 10, \beta = 1$). The VQ-VAE is trained for 50 epochs with identical lr/decay, using a `StepLR` scheduler (step=15, $\gamma = 0.5$) and loss weights $\alpha = 25, \beta = 1$. Codebook utilization reaches 128/128 active slots within the first epochs, confirming EMA stability. The Temporal Transformer and token-level prior use $\text{lr}=10^{-4}$ and gradient clipping at 1.0. The frame-level prior is optimized via MSE for 30 epochs on the 50k dataset.

E. Look-ahead labelling. A binary target $y_t \in \{0, 1\}$ is extracted via the look-ahead boundary condition:

$$y_t = \mathbb{I}(\exists \tau \in (t, t + 15] \text{ s.t. } \Delta \text{Lives}_\tau < 0) \quad (1)$$

To mitigate the 94%/6% class imbalance, all single-frame classifiers draw training batches through a class-balanced `WeightedRandomSampler`. The continuous baseline additionally weights its BCE loss by the empirical inverse-frequency ratio (15:1), while the VQ-VAE uses a milder 3:1 cross-entropy weight, since the sampler already balances each batch.

F. Reproducibility. The execution environment is containerized. Code, pre-trained weights, configurations, and both compiled datasets (10k and 50k) are hosted on Kaggle and accessible via <https://github.com/leonardoCosta02/pacman-world-model>.